1. Probability & Set Theory

Quick Exercise

1. Roll a fair six-sided die ($\Omega = \{1, 2, 3, 4, 5, 6\}$). Find $P(\text{roll} < 4)$.

---

2. Operations with Events

Quick Exercise

1. If $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$, find:
   - $P(A^c)$
   - $P(A \cup B)$
   - $P((A \cup B)^c)$

---

3. Conditional Probability & Independence

Quick Exercise

1. Two fair coin tosses: Show that "First toss = Heads" and "Second toss = Heads" are independent.

---

5. Discrete vs. Continuous RV Examples

Quick Exercise

1. A fair coin is tossed 5 times. Let X be the number of heads. Find $P(X = 2)$ and $E[X]$.

---

6. Normal Distribution & Applications

Quick Exercise

1. If $X \sim N(100, 16)$, find $P(90 \leq X \leq 110)$ using Z-scores or tables.

---

7. Basic Statistical Inference

Quick Exercise

1. You sample 100 people's heights. Which is the parameter, and which is the statistic?

---

8. Point Estimation & Confidence Intervals

Quick Exercise

1. A sample of size $n = 25$ from a normal population has $\bar{X} = 50$ and $s = 4$. Construct a 95% confidence interval for μ.